CIT 308 – Data Visualization

Tutorial 01

Question 01

1. What is Data Visualization?
The act of displaying data in a graphical or visual format to help in the understanding of correlations, patterns, trends, and insights is known as data visualization. Visualization facilitates rapid comprehension of complicated information rather than relying solely on text-based data or raw figures.

2. Name three real-world applications of data visualization.
Business Analytics
Healthcare
Finance

3. What is the difference between quantitative and qualitative data in visualization?
Quantitative
Quantitative data is information that can be measured and expressed in numbers. counts, or values, sales data, or temperature measurements. Charts such as scatter plots, bar charts, histograms, and line graphs are used in data visualization to display quantitative data and highlight patterns, trends, and comparisons
Qualitative

Non-numerical information that characterizes groups, attributes, or traits is known as qualitative data. Product categories, consumer reviews, hues, and geographical areas are a few examples. Pie charts, bar charts, and word clouds are frequently used in visualizations of qualitative data to illustrate proportions, frequency, or distinctions across categories.

4. Which type of chart would you use for showing:
a) Relationship between two variables? **Scatter Plot**
b) Distribution of exam marks? **Histogram**
c) Comparison of sales across products? **Bar Chart**

Question 02

1. Explain the three main types of data with examples.

Quantitative Data – Data that can be measured and expressed in numbers.
Example: Temperature readings, number of students, monthly sales figures.

Qualitative Data – Data that describes categories, qualities, or characteristics.
Example: Product types, customer feedback, colors, or regions.
Categorical Data – A subtype of qualitative data where values belong to specific groups or categories.
Example: Blood group (A, B, AB, O), education level (high school, bachelor's, master's).

2. Explain how structured, semi-structured, and unstructured data would be used in building a crime analysis dashboard.

Structured data is easier to find and analyse since it is well-organised and kept in tables with rows and columns. incident records containing elements including date, location, kind of incident, and suspect details are examples of structured data that may be used in a crime analysis dashboard. Charts, graphs, and heat maps displaying crime trends, hotspots, or regional comparisons may be made straight from this data.

3. Identify whether the following are Quantitative or Qualitative:

i) Blood group → **Qualitative**
ii) Height in cm → **Quantitative**
iii) Movie genres → **Qualitative**
iv) Marks in an exam → **Quantitative**

4. Consider a dataset of COVID-19 cases(Date, Country, Number of Cases). Which type of visualization is best to show:

a) Trend of cases over time?

A line chart is ideal for showing the trend of cases over time. The x-axis would represent the Date, and the y-axis would represent the Number of Cases, making it easy to see how the case numbers have risen or fallen.

b) Comparison of cases between countries?

A bar chart is the most effective way to compare cases between different countries. Each bar would represent a country, and the length of the bar would correspond to the Number of Cases, allowing for a clear side-by-side comparison

c) Distribution of cases in a single country?

A histogram is best for visualizing the distribution of cases within a single country. It can show the frequency of different case counts, helping to identify common ranges or outliers in the data.

Question 03

1. **Write a Python program to:**
   - **Store 5 student names and marks in a dictionary.**
   - **Print all students who scored above 75.**

Student Name	Marks
Alice	90
Bob	65
Charlie	80
Diana	72
Even	88

```
student_marks = {
  "Alice": 90,
  "Bob": 65,
  "Charlie": 80,
  "Diana": 72,
  "Even": 88
}
print("Students who scored above 75:")

# Iterate through the dictionary and print students with marks > 75
for student, mark in student_marks.items():
  if mark > 75:
    print(student)
```

2. **What is the difference between:**
   - **List and Tuple**
     - **List:** A list is a mutable sequence, meaning its elements can be added, removed, or changed after creation. It is defined using square brackets [].
     - **Tuple:** A tuple is an immutable sequence, meaning its elements cannot be changed after creation. It is defined using parentheses ().

- **Set and Dictionary**
  - **Set:** An unordered collection of unique elements. It is used to store multiple items in a single variable and is defined using curly braces {}. A set only stores values.
  - **Dictionary:** An unordered collection of key-value pairs. Each item in a dictionary has a key that is used to access its corresponding value. It is also defined using curly braces {}.

3. Write Python code to create:

a) A list of three fruits.

```
fruits = ["apple", "banana", "cherry"]
print("List of fruits:", fruits)
```

b) A tuple of three cities.

```
cities = ("New York", "Paris", "Tokyo")
print("Tuple of cities:", cities)
```

c) A set of numbers from 1 to 5.

```
numbers = {1, 2, 3, 4, 5}
print("Set of numbers:", numbers)
```

d) A dictionary with student names and marks.

```
student_grades = {
    "John": 85,
    "Sarah": 92,
    "Mike": 78
}
print("Dictionary of student grades:", student_grades)
```

4. What is the output of the following?

```
a = 10
b = 3
print(a // b)
print(a % b)
print(a ** b)
```

```
output = 3
         1
         1000
```

5. Write Python code to display:

a) Current year

b) Current month

c) Current day

```
import datetime

# Get the current date and time

current_date = datetime.date.today()

# a) Display current year
print("Current year:", current_date.year)

# b) Display current month
print("Current month:", current_date.month)

# c) Display current day
print("Current day:", current_date.day)
```

Question 04

1. **Write a Python program to check if a person is eligible to vote (age ≥ 18).**

```
age = int(input("Enter your age: "))
if age >= 18:
   print("You are eligible to vote ")
else:
   print("You are not eligible to vote ")
```

2. **Write a program using a while loop to calculate the sum of numbers from 1 to 10**

```
i = 1
total = 0
while i <= 10:
   total += i
   i += 1
print("Sum of numbers from 1 to 10 is:", total)
```

3. **Write a program using a for loop to print all elements of a list.**

```
my_list = ["apple", "banana", "cherry", "date"]

print("Printing elements of the list:")

for item in my_list:
   print(item)
```

4. **Write a Python program to store 5 students' names and marks in a dictionary, and print only those who scored above 75.**

```
student_marks = {
  "Alice": 90,
  "Bob": 65,
  "Charlie": 80,
  "Diana": 72,
  "Even": 88
}

print("Students who scored above 75:")

for student, mark in student_marks.items():

  if mark > 75:
    print(student)
```

5. **Write a Python program that:**
   • Asks the user for a number.
   • Prints whether it is prime or not prime.

```
num = int(input("Enter a number: "))
if num <= 1:
  print("Not prime ")
else:
  is_prime = True
  for i in range(2, int(num**0.5) + 1):
    if num % i == 0:
      is_prime = False
      breakif is_prime:
    print("Prime ")
  else:
    print("Not prime ")
```

6. **Consider the dataset:**
   [12, 45, 23, 67, 45, 89, 12, 45, 23]
   Write a program to:
   a) Find the unique values.
   b) Count how many times each number appears

```
data = [12, 45, 23, 67, 45, 89, 12, 45, 23]

# a) Find the unique values
unique_values = list(set(data))
print("Unique values:", unique_values)
```

```
# b) Count how many times each number appears
from collections import Counter
counts = Counter(data)
print("Count of each number:", counts)
```